

The Edge Negotiator

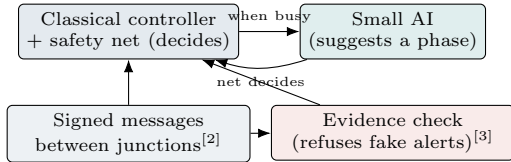
Verified-Source Cross-Junction Coordination for SLM-Driven Traffic Signal Control

Progress report · UCL MSc Systems Engineering for IoT (25153651) · Supervisors: Dr A. Delibasi (UCL), L. Stott (Microsoft) · 2026-06-20

1. What the project is

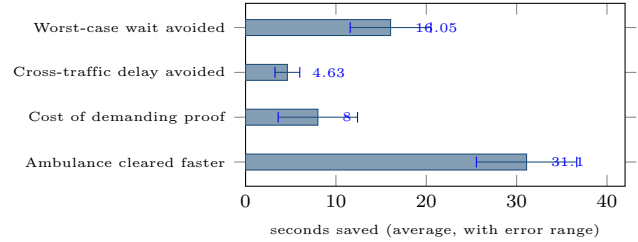
A row of traffic junctions, each run by a small AI model (**Phi-4-mini**, running locally on **Microsoft Foundry Local**, never retrained). The junctions tell each other what they are about to do over a channel that is digitally signed and sanity-checked, on top of a fast classical controller (*MaxPressure*^[1]) that stays in charge by default. A **deterministic safety net always makes the final, safety-critical call**. The real goal is **graceful failure under a lying neighbour**: if a junction's messages are tampered with, or the junction itself is hijacked, the system notices the story does not add up and falls back safely instead of believing it. In this demo, clearing a real ambulance and refusing a faked one is done by that deterministic safety net: it grants priority only with real physical evidence the vehicle exists, not on a valid signature alone. The AI assists with ordinary coordination today; its distinctive job, reasoning over genuinely ambiguous multi-junction situations, is what we evaluate next.

2. How it works



The classical controller runs things, with the AI suggesting a phase when busy. A deterministic safety net makes the final call. Every neighbour message is signed; a faked emergency fails the evidence check and is refused.

3. Main result (measured over 30 runs)



Our system vs two baselines on identical traffic (SUMO^[4] simulator). All four effects are clearly above zero.

4. What we have done

- **Built and tested:** signed messaging between junctions, an approved-senders list, the classical controller with its safety net, a check that flags neighbour reports that do not match the traffic actually on the road, and a four-junction simulated corridor.
- **Emergency handling:** a junction clears an ambulance only when it has real evidence the vehicle exists, not just a valid signature, so a faked emergency is refused.
- **Live attack-and-defence demo:** a hijacked junction sends a signed but fake “ambulance coming” alert and it is refused. A real ambulance is cleared at every junction. Measured against

a system that trusts everything and against the plain classical controller.

- **Evidence (30 runs):** emergency handling clears the ambulance **31 s faster** than having none; the defence blocks an attack that would otherwise add **16 s** to the worst side-street wait; insisting on evidence costs **8 s** versus blindly trusting, a deliberate and measured trade-off.
- **Independent check:** an automated audit of our written specification found *no made-up sources*; a few over-stated citations were corrected. Every result can be reproduced from its settings.

5. Why it is new

No existing system has been tested as a multi-junction, AI-coordinated traffic controller where messages between junctions can be *faked by an authenticated but hijacked junction*. Each ingredient exists alone; the combination does not. Closest work: **CoLLMLight**^[5] (AI junctions that coordinate but trust each other completely), **VLMLight**^[6] (fast-classical plus slow-AI, single junction, very large models, no trust), **LA-Light**^[7] (AI for rare events, cloud-based, no malicious faults), **Keijzer 2021**^[8] (detects coordination attacks, but only in theory). Our difference: a small on-device AI, a *measured* picture of how well detection holds up as the attack grows, verifiable identity, and a tamper-proof log.

6. Next steps

- Bigger evaluation: more traffic levels and a real London (Lambeth) road, many repeated runs with error ranges, extending the 30-run result we already have.
- Find the breaking point: grow the size of the lie until detection fails, and report exactly where that happens.
- Heavier traffic, where both the ambulance benefit and the at-

tack damage are largest.

- Run the real Phi-4-mini on Foundry Local and confirm it answers consistently.
- Separate the cost of demanding evidence from the rest of the policy (a clean side-by-side).

7. What we do and do not claim

We claim: trustworthy messaging between junctions; reliable detection of faked or faulty neighbour reports, with a measured reliability curve; a guaranteed safety floor; and a fair attack-and-defence test. **We do not claim:** new cryptography; that the AI beats the classical controller on ordinary traffic; that the on-device AI yet improves the emergency decision (it is deterministic today; the AI's value on ambiguous cases is evaluated next); defence against two or more junctions colluding, or a stolen master key; or that simulator results transfer straight to real hardware (future work).

References. [1] Varaiya, MaxPressure control, *Transp. Res. C* 36 (2013). [2] Bernstein et al., Ed25519 signatures (2012); approved-sender registry. [3] Lighthill & Whitham, *Proc. R. Soc. A* 229 (1955); Richards, *Oper. Res.* 4 (1956): the traffic conservation law behind the sanity check. [4] Lopez et al., Microscopic Traffic Simulation using SUMO, *IEEE ITSC* (2018). [5] Yuan et al., CoLLMLight, arXiv:2503.11739 (2025). [6] Wang et al., VLMLight, arXiv:2505.19486 (2025). [7] Wang et al., LA-Light, arXiv:2403.08337 (2024). [8] Keijzer et al., Detection of Cyber-Attacks in Collaborative Intersection Control, *ECC* (2021).